Learner Full Name: _____

School: _____ **Class:** _____

Workstation Device: _____ **Station #:** _____

Session 10: Spreadsheets, Tables & Basic Formulas

In Level 4, you learned how to use a word processor to write essays and notices. Today, we unlock a completely different digital tool designed to handle **numbers, lists, money, and calculations**: the **Spreadsheet**!

1. Plain Talk: The Market Trader's Ledger Book

If you visit a busy shop or stall at Makola Market or Kejetia Market, the shopkeeper keeps a thick paper **ledger book** ruled into rows and columns. They write down items sold, quantities, and prices. At the end of the day, they use a pen or calculator to add up the cash totals line by line.

An electronic **Spreadsheet** (like Microsoft Excel, Google Sheets, or LibreOffice Calc) is an intelligent digital ledger. When you change a single price or quantity, **the spreadsheet recalculates all totals automatically in less than a millisecond!**

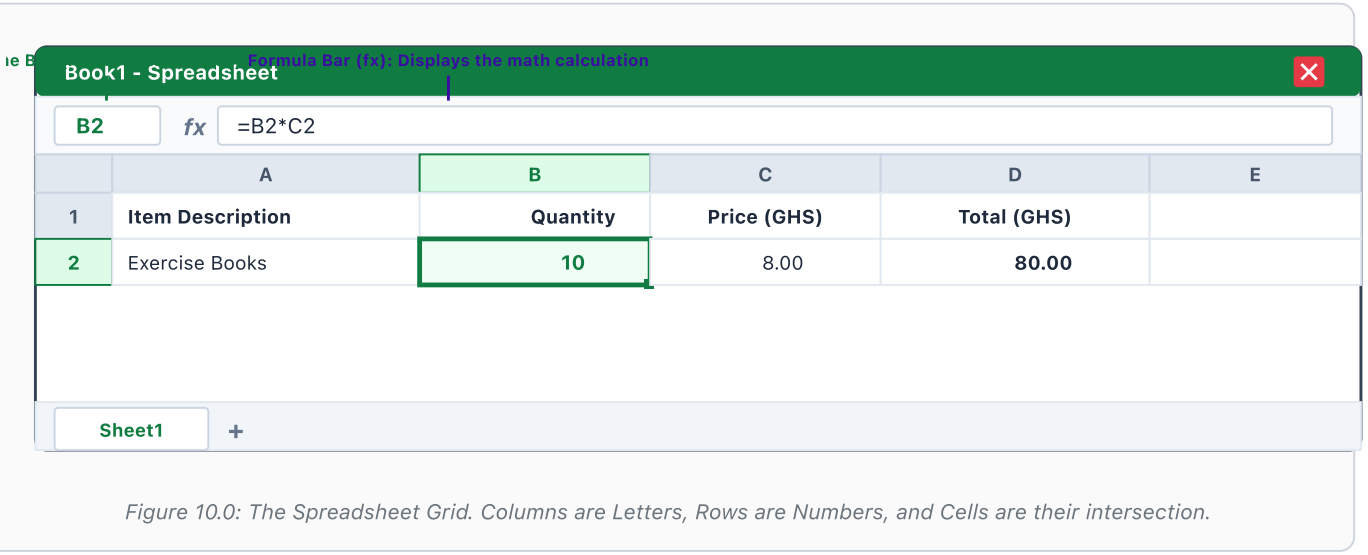


Figure 10.0: The Spreadsheet Grid. Columns are Letters, Rows are Numbers, and Cells are their intersection.

2. Essential Spreadsheet Vocabulary

- **Column:** A vertical strip of boxes identified by a **Letter** (A, B, C...).
- **Row:** A horizontal line of boxes identified by a **Number** (1, 2, 3...).
- **Cell:** A single box in the grid formed where a Column meets a Row (e.g., Cell B2).
- **Active Cell:** The box currently selected with a dark green border around it.
- **Formula Bar (fx):** The long white box at the top where formulas and raw data are displayed.

1. Cell Coordinate Addresses (Column Letter + Row Number)

Every single box in a spreadsheet has its own unique home address. You find the address by reading the **Column Letter first**, then the **Row Number second**:

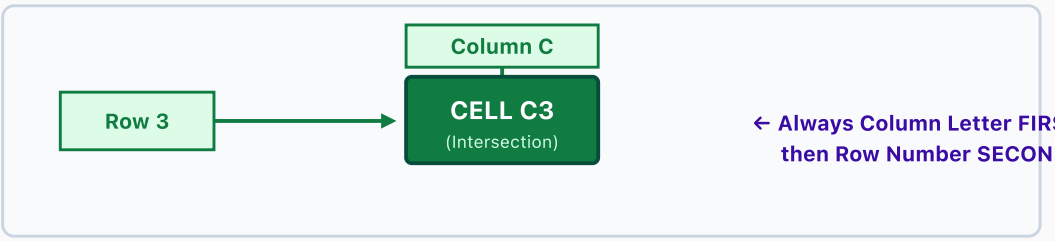


Figure 10.1: Reading cell coordinates. Column C meets Row 3 at Cell C3.

2. The Two Basic Types of Data

🔤 TEXT (Labels)

Words, names, descriptions, or titles. Text helps people understand what the numbers mean.

Examples: Tomatoes, Ama Mensah, Exercise Books

Alignment: The computer automatically hugs text to the **LEFT** side of the cell.

🔢 NUMBERS (Values)

Raw digits, quantities, prices, percentages, and measurements that can be mathematically calculated.

Examples: 10, 25.50, 100

Alignment: The computer automatically pushes numbers to the **RIGHT** side of the cell!

3. Check Your Thinking

Look at the coordinate grid below. Write the cell address for each item:

	A	B	C
1	Rice	Gari	Beans
2	Yam	Plantain	Cassava

A. In which cell is **Gari** located? _____

B. In which cell is **Cassava** located? _____

THE GOLDEN LAW OF FORMULAS

1. What is a Formula?

A **Formula** is an electronic mathematical instruction you give to a cell. Instead of doing the math in your head and typing the answer, you write a formula that commands the computer to calculate it for you!

> 👑 **THE GOLDEN LAW OF SPREADSHEETS:**
>
> **EVERY FORMULA MUST START WITH AN EQUALS SIGN: =**
>
> If you forget to type the **=** sign first, the computer thinks you are just typing plain words and will **NOT** calculate anything!

2. The Four Computer Math Operators

Computers do not use the math symbols from your primary school exercise notebook. Study these keyboard operators:

Math Operation	Old Paper Symbol	Computer Keyboard Key	How to Write It in a Cell
Addition	+	+ (Shift + =)	=B2 + C2
Subtraction	–	- (Minus key)	=B2 - C2
Multiplication	×	* (Asterisk: Shift + 8)	=B2 * C2 *(Never type the letter 'x'!)*
Division	÷	/ (Forward slash)	=B2 / C2

3. Why We Use Cell Addresses (Not Fixed Numbers!)

❌ **BAD FORMULA: =10 * 8**

If you type =10 * 8, the cell shows 80. But if the store manager changes the price to 9 Cedis tomorrow, your total stays stuck at 80! It is dead math.

✅ **SMART FORMULA: =B2 * C2**

You tell the cell: *"Multiply whatever number is in B2 by whatever is in C2!"* If prices change next week, the total updates automatically! This is called **Dynamic Calculation**.

1. What is the AutoFill Handle?

Imagine having a sales list of 200 items in a school store. You do **not** have to type =B3*C3, =B4*C4, =B5*C5 two hundred times!

When you select a cell, look at its bottom-right corner. There is a tiny dark green square called the **AutoFill Handle**. When you drag it down, the computer copies the math formula and updates the row numbers automatically!

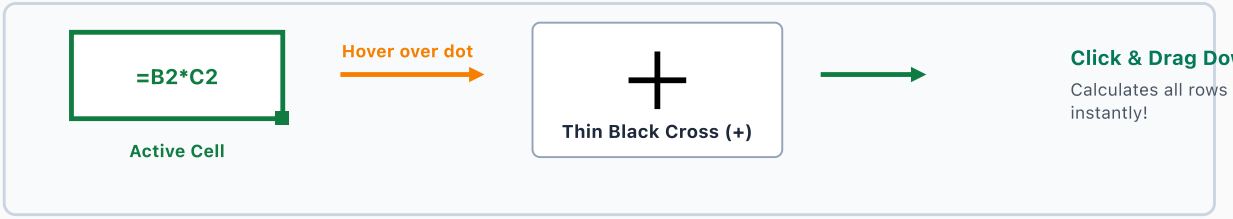


Figure 10.2: The AutoFill Handle. When the cursor turns into a thin black cross (+), drag down to copy formulas.

2. Panic Button: Spreadsheet Error Messages Decoded

Error on Screen	What Actually Happened	How to Fix It in 2 Seconds
##### (Train Tracks)	The number is too wide to fit inside the cell column width!	Move your pointer to the line between column letters (e.g. between D and E) and **double-click** to widen it!
#NAME?	You misspelled a formula name or forgot the equals sign.	Click the cell and check the Formula Bar. Make sure it starts with =.
#VALUE!	You tried to do math with words (e.g. multiplying "Books" * 10).	Check your formula: make sure it points only to cells with numbers inside.

1. Lab Mission: Build a 4-Item Inventory Table

Follow these 5 steps to create your first working spreadsheet in Excel, Calc, or Google Sheets:

1 **Launch Spreadsheet Software:**
Click Start → type excel (or open **LibreOffice Calc** or **Google Sheets**) → Open a blank workbook.

2 **Type Column Headers in Row 1:**
- In Cell **A1**, type: Item Description → tap Tab
- In Cell **B1**, type: Quantity → tap Tab
- In Cell **C1**, type: Unit Price (GHS) → tap Tab
- In Cell **D1**, type: Line Total (GHS) → tap Enter

3 **Enter the Stationery Items (Rows 2 to 5):**
- Row 2: Exercise Books | 10 | 8
- Row 3: Ballpoint Pens | 20 | 2
- Row 4: Wooden Rulers | 5 | 4
- Row 5: Pencil Erasers | 15 | 1.5

4 **Write the Multiplication Formula:**
Click inside Cell **D2**. Type: =B2*C2 and press `Enter`.
Observation: Cell D2 instantly calculates **80**!

5 **AutoFill Down & Save:**
Click Cell **D2** again. Hover over the bottom-right green dot until the cursor turns into a thin black cross (**+**).
Click and drag down to **D5**.
All line totals appear automatically!
Save: **File → Save As** → Documents > BIH_Practice → Name: Stationery_Store_Ama.xlsx.

2. Visual Target (What Your Finished Sheet Must Look Like)

	A	B	C	D
1	**Item Description**	**Quantity**	**Unit Price (GHS)**	**Line Total (GHS)**
2	Exercise Books	10	8.00	**80.00**
3	Ballpoint Pens	20	2.00	**40.00**
4	Wooden Rulers	5	4.00	**20.00**
5	Pencil Erasers	15	1.50	**22.50**

SESSION 10 REVIEW & VERIFICATION

1. Formula Writing Practice (On Paper)

Write the correct spreadsheet formula you would type into the destination cell:

Math Problem to Solve	Your Formula (Remember the Golden Law!)
Multiply Cell **B3** by Cell **C3**:	
Add Cell **A1** and Cell **A2** together:	
Divide Cell **D10** by Cell **E10**:	
Subtract Cell **C5** from Cell **B5**:	

2. Written Review Questions

Question 1: Why does a spreadsheet automatically push numbers to the right side of a cell, but words to the left side?

Question 2: A student types B2*C2 into a cell and presses Enter. The cell just displays the text "B2*C2" instead of calculating a number. What critical mistake did they make?

Question 3: You see ##### filling an entire cell. What does this mean, and how do you fix it?

3. Self-Assessment & Mentor Sign-Off

Tick the column that reflects your confidence at the end of Session 10:

Spreadsheet Competency	With Help	Almost Alone	All By Myself!
1. I can locate any cell using Column Letters and Row Numbers (e.g. B2).	○	○	○
2. I remember to start every single math formula with an equals sign (=).	○	○	○
3. I can write multiplication formulas using the asterisk key (*).	○	○	○
4. I can use the AutoFill handle (+) to copy formulas down a column.	○	○	○

Mentor Signature: _____ **Date Completed:** _____